

Introduction to Kotlin

kotlinlang.org/education



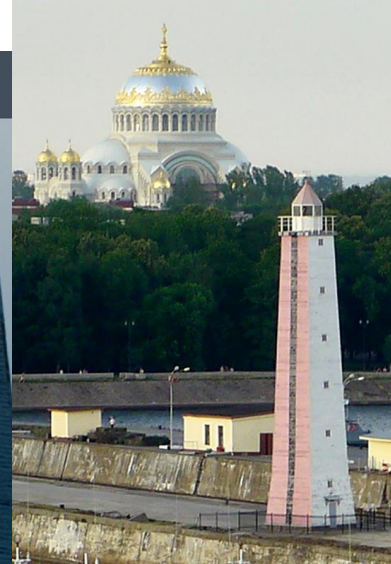
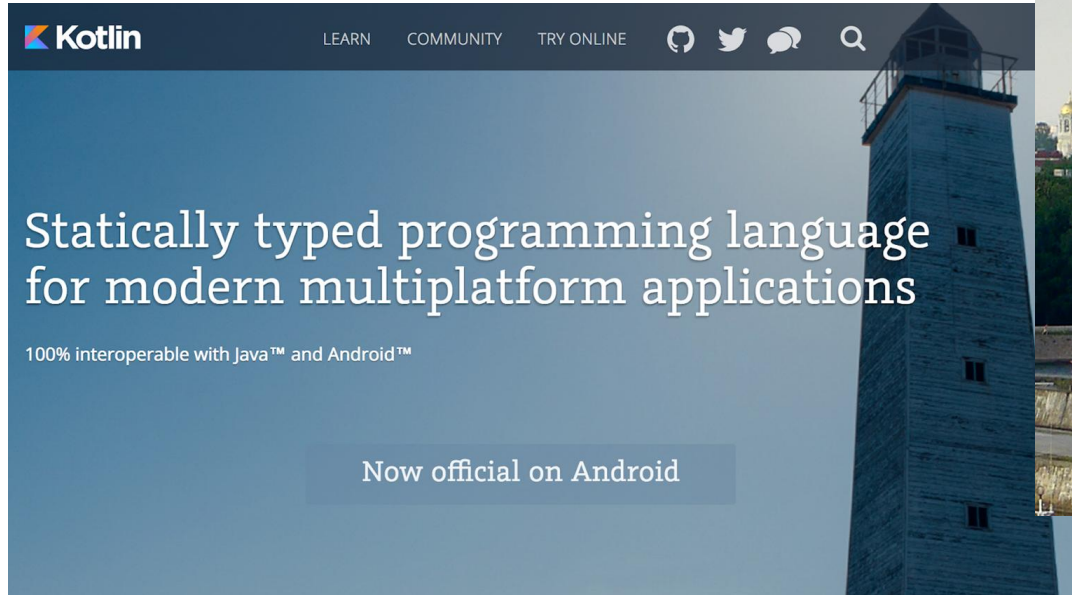
Why Kotlin?

- Expressiveness/Conciseness
- Safety
- Portability/Compatibility
- Convenience
- High Quality IDE Support
- Community
- Android 🧑🏻🧑🏻
- More than a gazillion devices run ~~Java~~ Kotlin
- Lactose free
- ~~Sugar free~~
- Gluten free

Logo



Name



Kotlin is named after an island in the Gulf of Finland.

Hello, world!

```
fun main(args: Array<String>) {  
    println("Hello, world!")  
}
```

```
fun main() {  
    println("Hello, world!")  
}
```

```
fun main() = println("Hello, world!")
```

Where is ";"???

The basics

```
fun main(args: Array<String>) {  
    print("Hello")  
    println(", world!")  
}
```

- An entry point of a Kotlin application is the `main` **top-level** function.
- It accepts a variable number of `String` arguments that can be omitted.
- `print` prints its argument to the standard output.
- `println` prints its arguments and adds a line break.

Variables

`val/var myValue: Type = someValue`

- `var` - mutable
- `val` - immutable
- Type can be inferred in most cases
- Assignment can be deferred

```
val a: Int = 1           // immediate assignment
```

```
var b = 2                // 'Int' type is inferred  
b = a                    // Reassigning to 'var' is okay
```

```
val c: Int               // Type required when no initializer is provided  
c = 3                    // Deferred assignment  
a = 4                    // Error: Val cannot be reassigned
```

Variables

```
const val/val myValue: Type = someValue
```

- `const val` - compile-time const value
- `val` - immutable value
- for `const val` use uppercase for naming

```
const val NAME = "Kotlin" // can be calculated at compile-time
```

```
val nameLowered = NAME.lowercase() // cannot be calculated at compile-time
```

Functions

```
fun sum(a: Int, b: Int): Int {  
    return a + b  
}
```

```
fun mul(a: Int, b: Int) = a * b
```

```
fun printMul(a: Int, b: Int): Unit {  
    println(mul(a, b))  
}
```

```
fun printMul1(a: Int = 1, b: Int) {  
    println(mul(a, b))  
}
```

```
fun printMul2(a: Int, b: Int = 1) = println(mul(a, b))
```

Single expression function.

`Unit` means that the function does not return anything meaningful.

It can be omitted.

Arguments can have **default** values.

If expression

```
fun maxOf(a: Int, b: Int): Int {  
    if (a > b) {  
        return a  
    } else {  
        return b  
    }  
}
```

is the same as



```
fun maxOf(a: Int, b: Int) =  
    if (a > b) {  
        a  
    } else {  
        b  
    }
```

`if` can be an expression (it can return).

Can be a one-liner:

```
fun maxOf(a: Int, b: Int) = if (a > b) a else b
```

When expression

```
when (x) {  
  1 -> print("x == 1")  
  2 -> print("x == 2")  
  else -> {  
    print("x is neither 1 nor 2")  
  }  
}
```

`when` returns, the same way that `if` does.

```
when {  
  x < 0 -> print("x < 0")  
  x > 0 -> print("x > 0")  
  else -> {  
    print("x == 0")  
  }  
}
```

The condition can be inside of the branches.

When statement

```
fun serveTeaTo(customer: Customer) {  
    val teaSack = takeRandomTeaSack()  
  
    when (teaSack) {  
        is OolongSack -> error("We don't serve Chinese tea like $teaSack!")  
        in trialTeaSacks, teaSackBoughtLastNight ->  
            error("Are you insane?! We cannot serve uncertified tea!")  
    }  
  
    teaPackage.brew().serveTo(customer)  
}
```

`when` can accept several options in one branch. `else` branch can be omitted if `when` block is used as a *statement*.

&& vs and

`if (a && b) { ... }` VS `if (a and b) { ... }`

Unlike the `&&` operator, this function does not perform short-circuit evaluation.

The same behavior with OR:

`if (a || b) { ... }` VS `if (a or b) { ... }`

Loops

```
val items = listOf("apple", "banana", "kiwifruit")
```

```
for (item in items) {  
    println(item)  
}
```

```
for (index in items.indices) {  
    println("item at $index is ${items[index]}")  
}
```

```
for ((index, item) in items.withIndex()) {  
    println("item at $index is $item")  
}
```

Loops

```
val items = listOf("apple", "banana", "kiwifruit")
```

```
var index = 0
while (index < items.size) {
    println("item at $index is ${items[index]}")
    index++
}
```

```
var toComplete: Boolean
do {
    ...
    toComplete = ...
} while(toComplete)
```

The condition variable can be initialized inside to the `do..while` loop.

Loops

There are `break` and `continue` labels for loops:

```
myLabel@ for (item in items) {  
    for (anotherItem in otherItems) {  
        if (...) break@myLabel  
        else continue@myLabel  
    }  
}
```

Ranges

```
val x = 10
if (x in 1..10) {
    println("fits in range")
}
```

```
for (x in 1..5) {
    print(x)
}
```

```
for (x in 9 downTo 0 step 3) {
    print(x)
}
```

downTo and step are extension functions, not keywords.

'..' is actually T.rangeTo(that: T)

Null safety

```
val notNullText: String = "Definitely not null"
val nullableText1: String? = "Might be null"
val nullableText2: String? = null

fun funny(text: String?) {
    if (text != null)
        println(text)
    else
        println("Nothing to print :(")
}

fun funnier(text: String?) {
    val toPrint = text ?: "Nothing to print :("
    println(toPrint)
}
```

Elvis operator ? :

If the expression to the left of `?:` is not `null`, the Elvis operator returns it; otherwise, it returns the expression to the right.

Note that the expression on the right-hand side is evaluated only if the left-hand side is `null`.

```
fun loadInfoById(id: String): String? {  
    val item = findItem(id) ?: return null  
    return item.loadInfo() ?: throw  
    Exception("...")  
}
```

:-)

Safe Calls

`something?.otherThing` does not throw an NPE if `something` is `null`.

Safe calls are useful in chains. For example, an employee may be assigned to a department (or not). That department may in turn have another employee as a department head, who may or may not have a name, which we want to print:

```
fun printDepartmentHead(employee: Employee) {  
    println(employee.department?.head?.name)  
}
```

To print only for non-null values, you can use the safe call operator together with `let`:

```
employee.department?.head?.name?.let { println(it) }
```

Unsafe Calls

The not-null assertion operator (!!) converts any value to a non-null type and throws an **NPE** exception if the value is null.

```
fun printDepartmentHead(employee: Employee) {  
    println(employee.department!!.head!!.name!!)  
}
```

Please, avoid using unsafe calls!

TODO

Always throws a **NotImplementedError** at **run-time** if called, stating that operation is not implemented.

```
// Throws an error at run-time if calls this function, but compiles  
fun findItemOrNull(id: String): Item? = TODO("Find item $id")
```

```
// Does not compile at all  
fun findItemOrNull(id: String): Item? = { }
```

String templates and the string builder

```
val i = 10
```

```
val s = "Kotlin"
```

```
println("i = $i")
```

```
println("Length of $s is ${s.length}")
```

```
val sb = StringBuilder()
```

```
sb.append("Hello")
```

```
sb.append(", world!")
```

```
println(sb.toString())
```

Lambda expressions

```
val sum: (Int, Int) -> Int = { x: Int, y: Int -> x + y }  
val mul = { x: Int, y: Int -> x * y }
```

According to Kotlin convention, if the last parameter of a function is a function, then a lambda expression passed as the corresponding argument can be placed outside the parentheses:

```
val badProduct = items.fold(1, { acc, e -> acc * e })  
  
val goodProduct = items.fold(1) { acc, e -> acc * e }
```

If the lambda is the only argument, the parentheses can be omitted entirely (the documentation calls this feature "trailing lambda as a parameter"):

```
run({ println("Not Cool") })  
run { println("Very Cool") }
```

When in doubt

Go to:

- kotlinlang.org
- kotlinlang.org/docs
- play.kotlinlang.org/ByExample

Thank you

